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Biology Opinion Paper: Genetically Modified Crops: Risky or Beneficial?

Genetically modified (GM) crops have become a significant part of global agriculture over the past few decades. They are developed through gene editing or transfer techniques that allow scientists to enhance certain traits like pest resistance, drought tolerance, or nutritional content. Supporters argue that GMOs can help address food security and reduce pesticide use, especially in countries with rapidly growing populations or poor soil conditions (FDA). However, concerns remain about long-term health effects, environmental impacts such as loss of biodiversity, and the ethics of altering natural organisms. The regulation and acceptance of GM crops vary widely: for example, China is expanding GM corn planting to improve food output, while the European Union has historically placed strict limits on their use due to precautionary public health and environmental concerns (Chu and Cao; Niranjan).

The debate is sharply divided. Advocates of GM crops often cite improved yields, lower pesticide dependency, and enhanced nutrition as valid benefits. Golden Rice, for example, is fortified with vitamin A and has the potential to prevent blindness in developing countries (Bawa and Anilakumar). Additionally, GM crops like Bt corn can reduce the need for chemical insecticides. However, critics argue that GMOs may introduce allergens, contribute to antibiotic resistance, or lead to unforeseen environmental consequences such as "superweeds" or cross-contamination with wild plant species. Some claims—like GMOs causing cancer—have

been widely debunked (Ducharme), yet misinformation persists. Meanwhile, government authorities in regions such as the EU have cited long-term uncertainty and ecosystem disruption as valid reasons to regulate or restrict GMO use (Niranjan).

While GM crops can offer some benefits, I believe that we should be extremely cautious when altering what is natural. Countries like China are embracing GMOs to improve crop performance and food availability, but at the same time, the European Union's insistence on stricter regulation reflects a more balanced, safety-focused approach. This precautionary stance recognizes that the long-term impacts of GMOs on human health and the environment are not yet fully understood. Once introduced into ecosystems or the food supply, these modifications cannot be easily reversed. Enhancing what nature has already optimized may offer short-term gains, but it risks unintended biological consequences. Therefore, while GM crops may have a role in the future of agriculture, their development and use must be carefully regulated based on sound science and long-term sustainability.

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